

AoyuX Monthly Earthquake Forecasting Report for China Seismic Experimental Site (CSES)

version 1

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Monthly report for June 1, 2024 to June 30, 2024 (30 days)

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1. Earthquake Forecast for the Next 30 Days in the CSES

The earthquake catalog used in this report is updated as of May 31, 2024. The magnitude type used in the catalog is local magnitude (M_L) released by the China Earthquake Networks Center (CENC). The simulations to obtain the earthquake forecasts are performed with the Epidemic Type Aftershock Sequence (ETAS) model with spatially varying background rate (Nandan et al., 2021a; 2021b, 2022; Li et al., 2024).

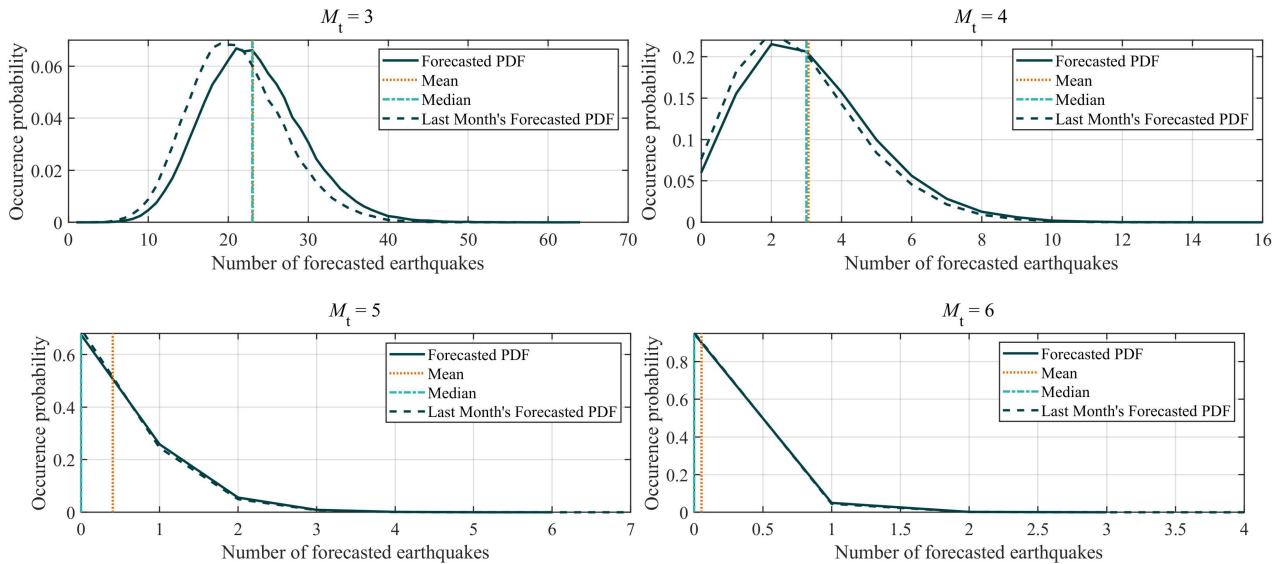


Figure 1. Forecasted probabilities for earthquakes of different magnitude ranges ($M \geq M_t$) for June 1, 2024 to June 30, 2024.

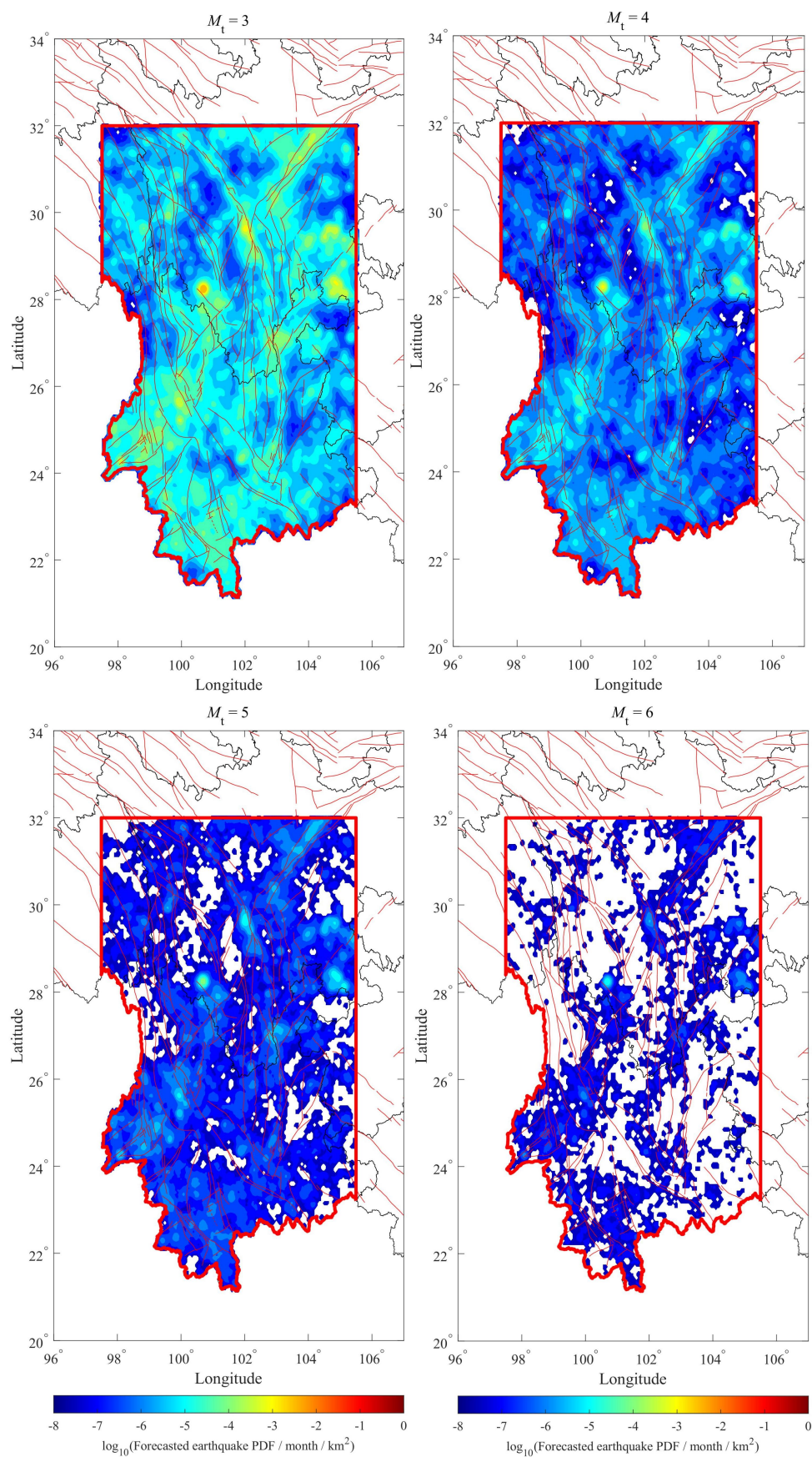


Figure 2. Probability density distribution (PDF) of forecasted earthquakes ($M \geq M_t$) for June 1, 2024 to June 30, 2024 per area (km^2).

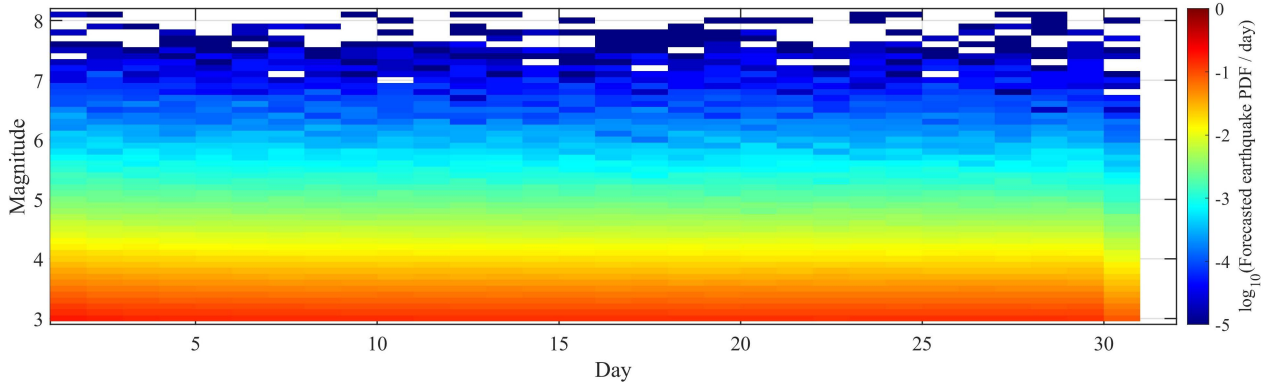


Figure 3. Daily probability density distribution (PDF) of forecasted earthquakes for June 1, 2024 to June 30, 2024.

2. Last Month's Forecast v.s. Realised Seismicity

In the past 31 days (May 1, 2024 to May 31, 2024), the region has experienced 32 earthquakes of M 3+, 5 of M 4+, and 2 of M 5+, with no earthquakes of M 6+.

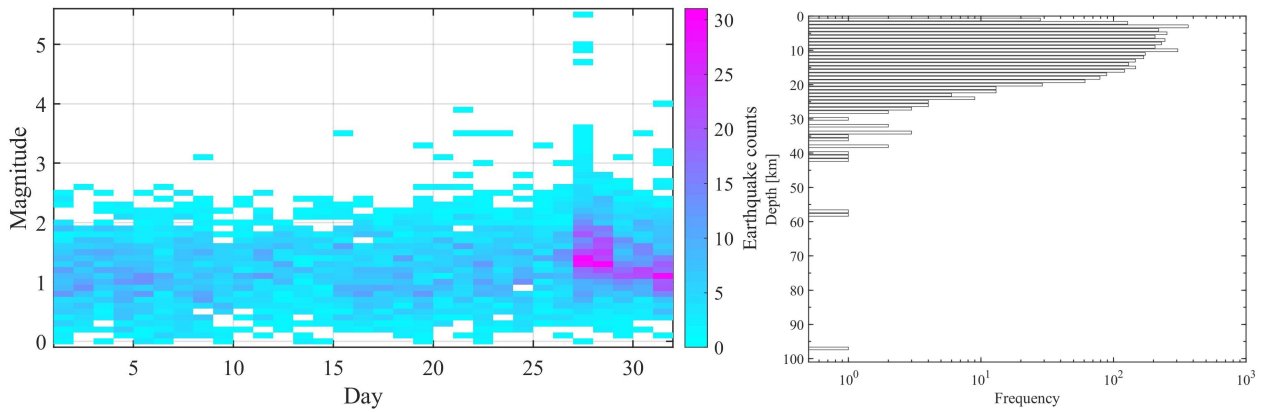


Figure 4. (left) Daily earthquake counts per 0.1 magnitude bin in the CSES over the past 31 days. (right) Depth-frequency distribution of earthquakes in the CSES over the past 31 days (May 1, 2024 to May 31, 2024).

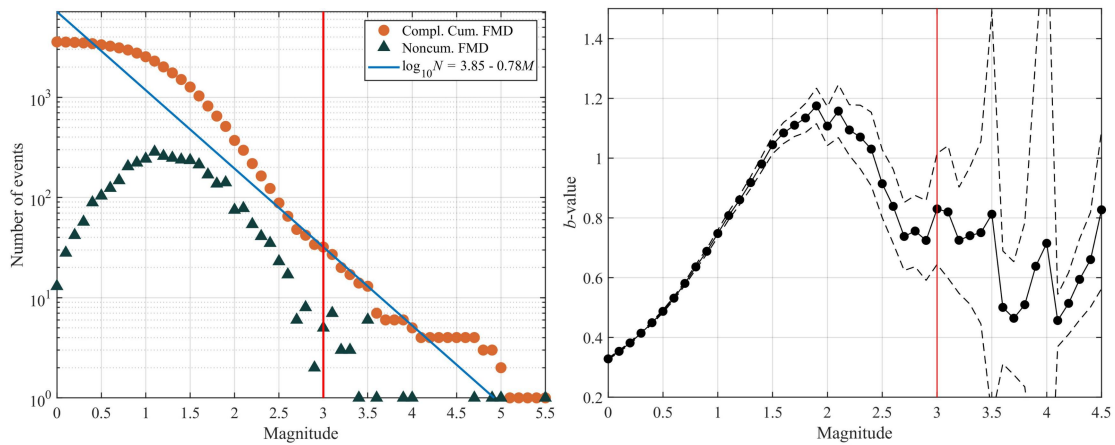


Figure 5. (left) Complementary cumulative and density frequency-magnitude distribution alongside the GR law fitted using earthquakes with magnitudes larger than the cut-off magnitude $M_{co} = 3.0$ (red line); (right) Variation of the b -value with M_{co} , with the thin dashed line indicating the $\pm 1\sigma$ standard deviation. The seismicity analyzed is from the past 31 days (May 1, 2024 to May 31, 2024).

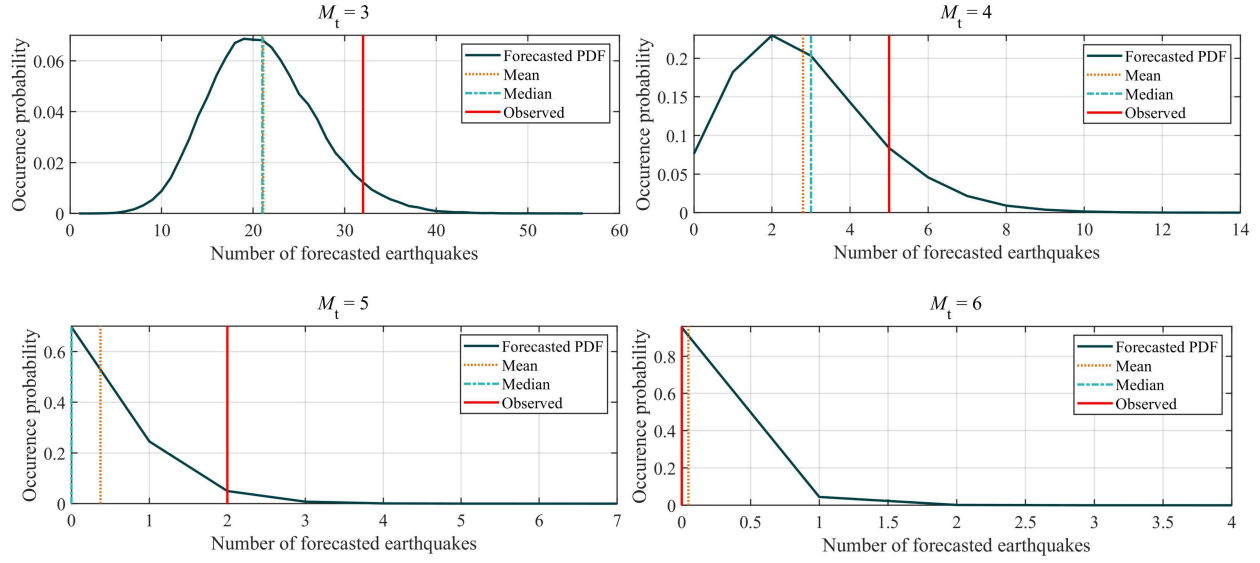


Figure 6. Number of observed earthquakes ($M \geq M_t$) in the CSES over the past 31 days (red vertical line) and forecasted probabilities for earthquakes of different magnitude ranges ($M \geq M_t$) published last month.

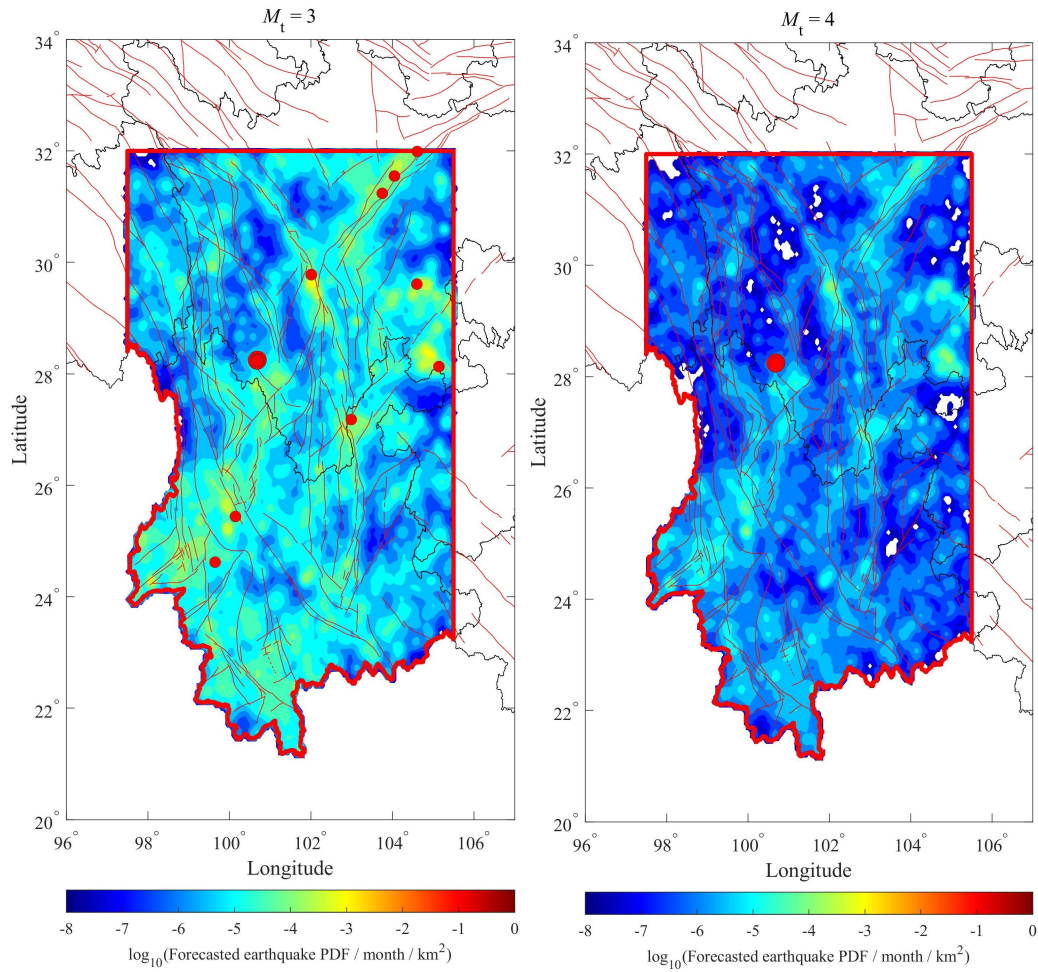


Figure 7. Observed seismicity (red solid circle) over the past 31 days (May 1, 2024, to May 31, 2024), with the probability density distribution (PDF) of forecasted earthquakes ($M \geq M_t$) for the same period per area (km^2), released by the last report as the background.

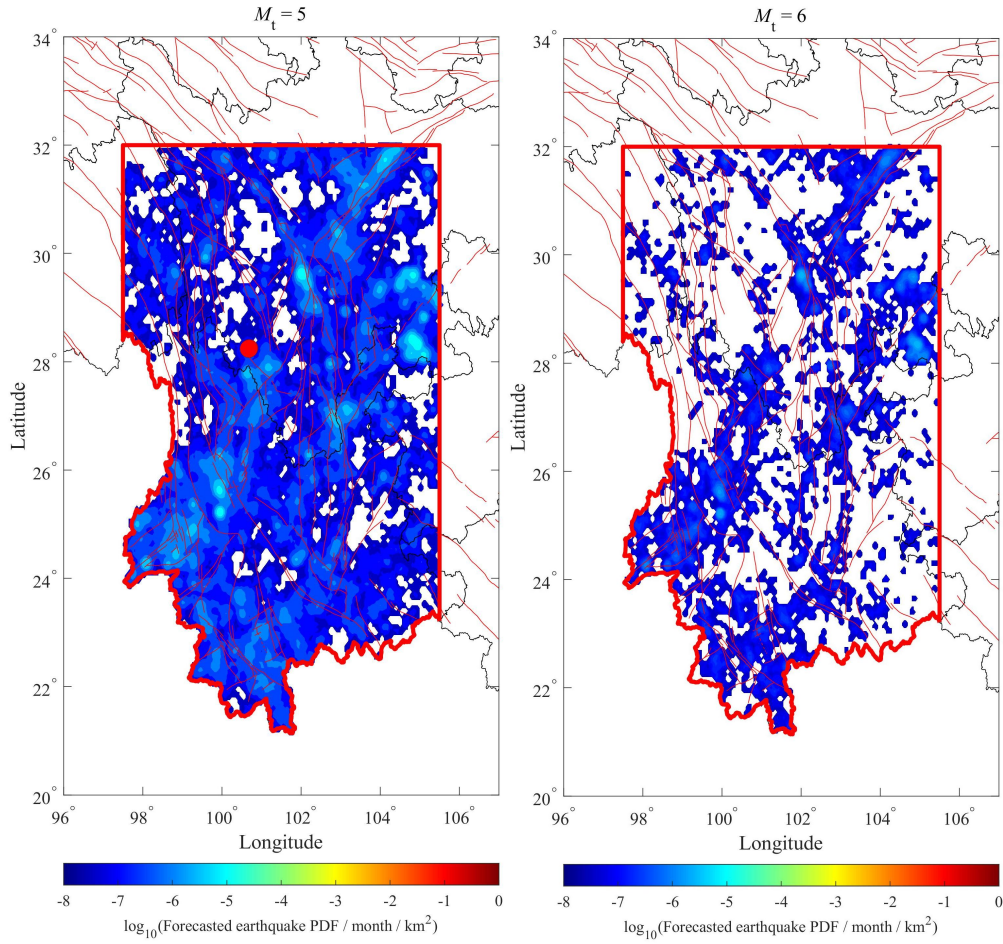


Figure 7. -continued.

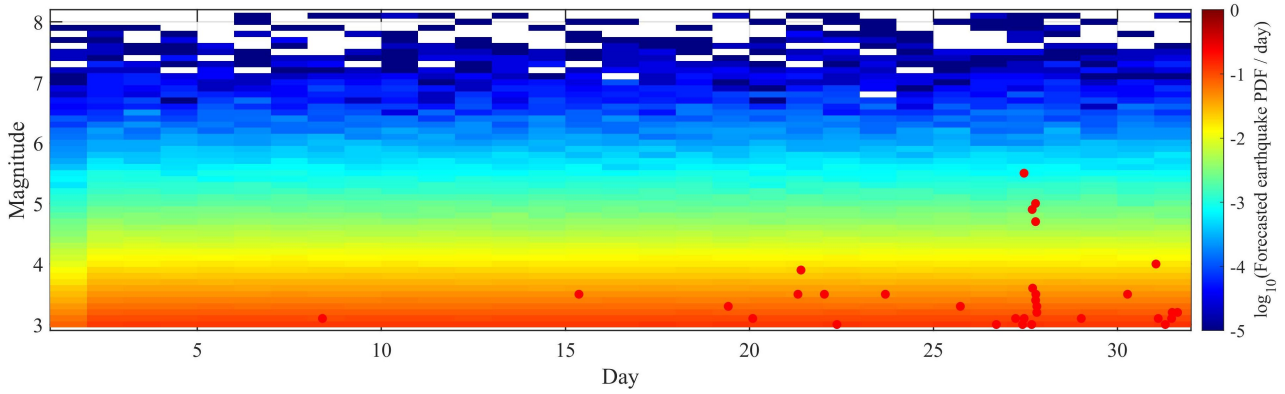


Figure 8. Observed seismicity (red solid circle) over the past 31 days (May 1, 2024, to May 31, 2024), with the daily probability density distribution (PDF) of forecasting earthquakes for the same period, released by last report as the background.

3. Background Information

Since the inception of seismic records at the CSES (~ 780 BC), there have been 210 earthquakes of $M \geq 6$, 43 of $M \geq 7$, and 2 of $M \geq 8$. Since 1970, there have been 534 earthquakes of $M \geq 5$, 91 of $M \geq 6$, 14 of $M \geq 7$, and 1 of $M \geq 8$.

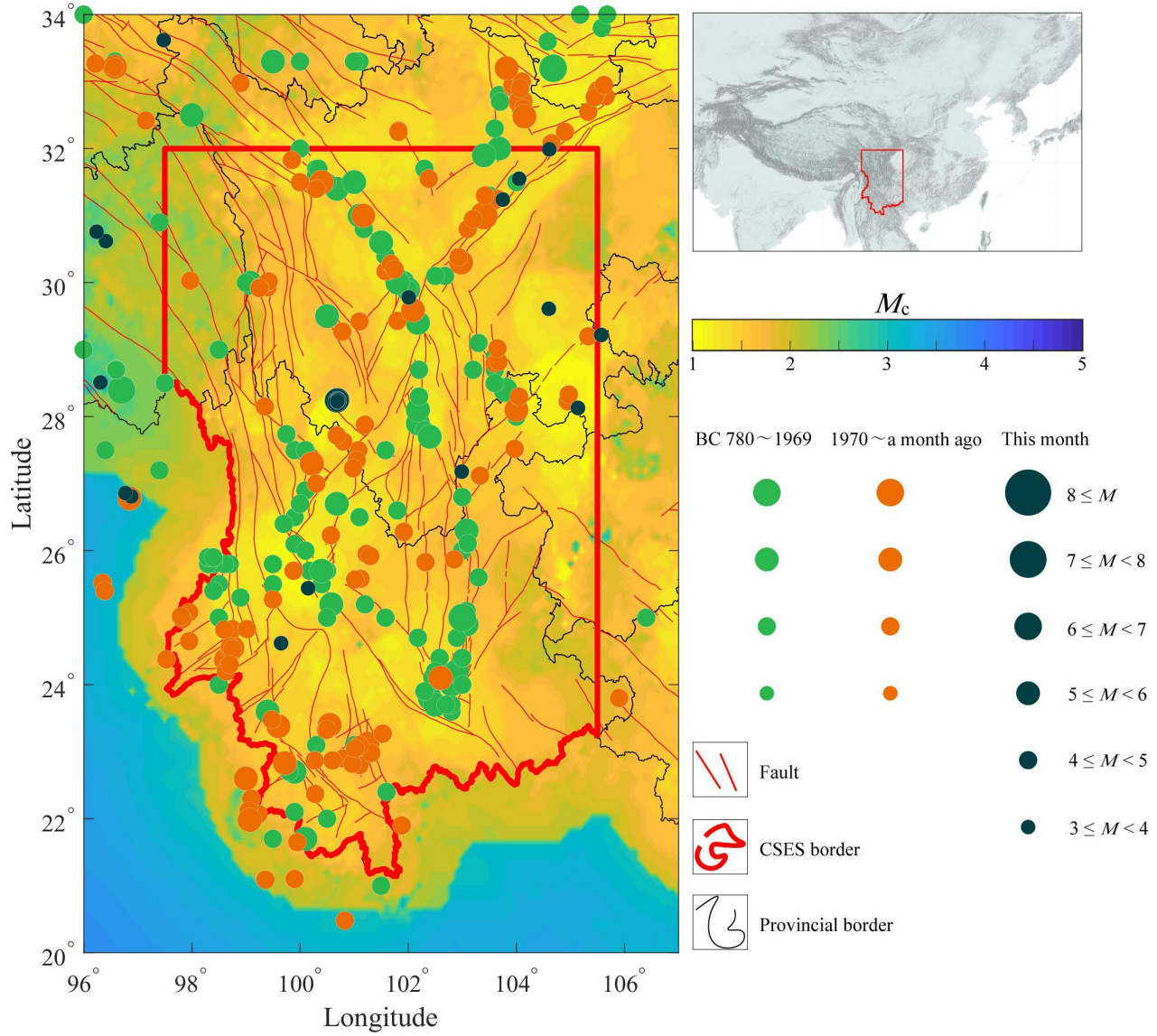


Figure 9. Spatial distribution of earthquakes and faults in the CSES located in the Sichuan-Yunnan region. The background map, showing the completeness magnitude (M_c), is adapted from Li et al. (2023).

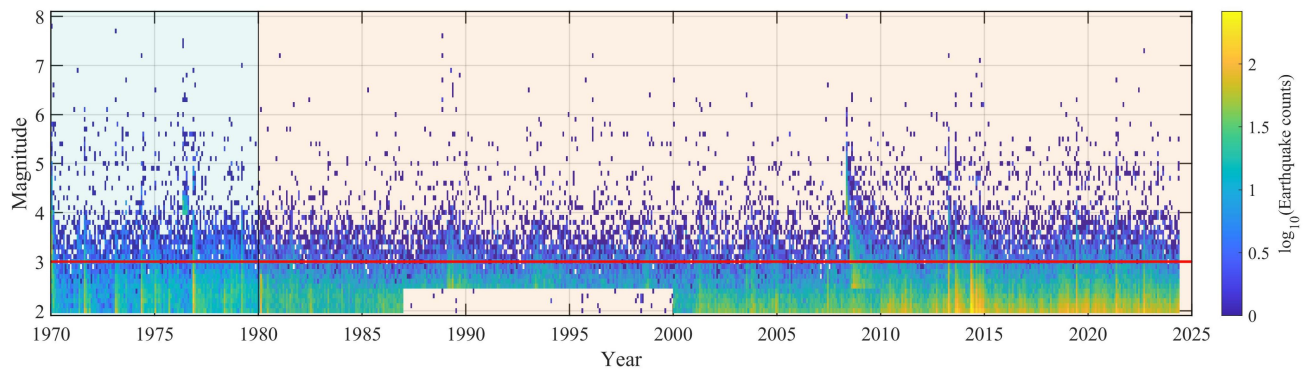


Figure 10. Monthly earthquake counts per 0.1 magnitude bin in the CSES. The red line indicates the cut-off magnitude $M_{co} = 3.0$. The light red and light blue shaded regions represent the primary period and the auxiliary time band, respectively. The parameters M_{co} , primary period, and auxiliary period are all utilized for the ETAS model calibration and simulations to build the forecasts.

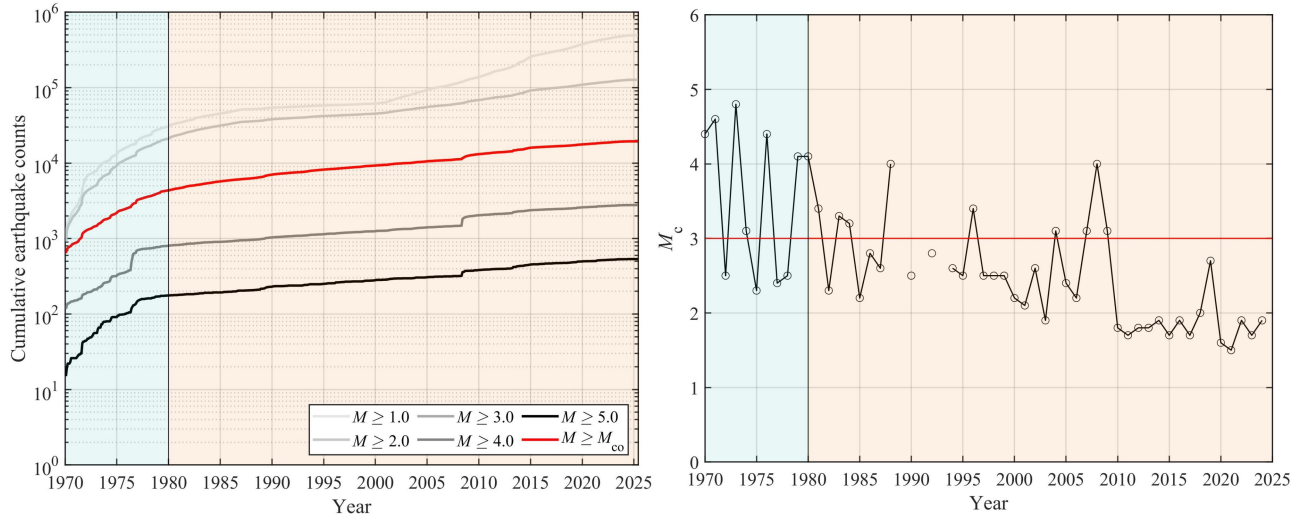


Figure 11. (left) Time series depicting cumulative earthquake counts for different cut-off magnitudes ($M_{co} = 1, 2, 3, 4,$ and 5) in the CSES; (right) Evolution of the completeness magnitude (M_c) over time. The method to estimate M_c is described in Li et al. (2024).

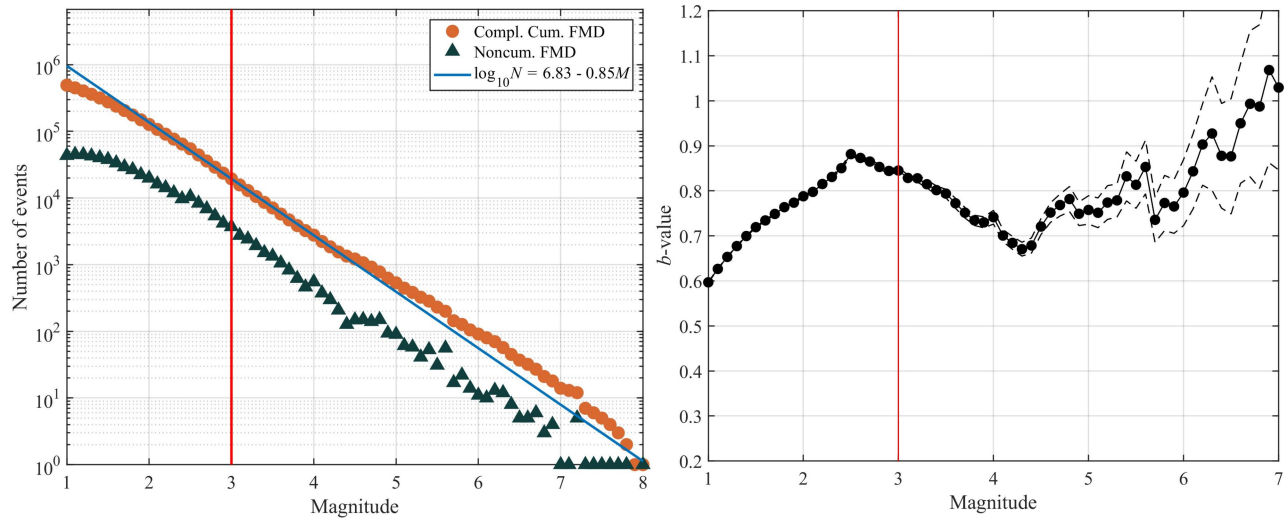


Figure 12. (left) Complementary cumulative and density frequency-magnitude distribution alongside the Gutenberg-Richter (GR) law fitted using earthquakes with magnitudes larger than $M_{co} = 3.0$ for the CSES catalogue since 1970; (right) Variation of the b -value with M_{co} , with the thin dashed line indicating the $\pm 1\sigma$ standard deviation. The methods to estimate b -value and fit the GR law is described in Li et al. (2024).

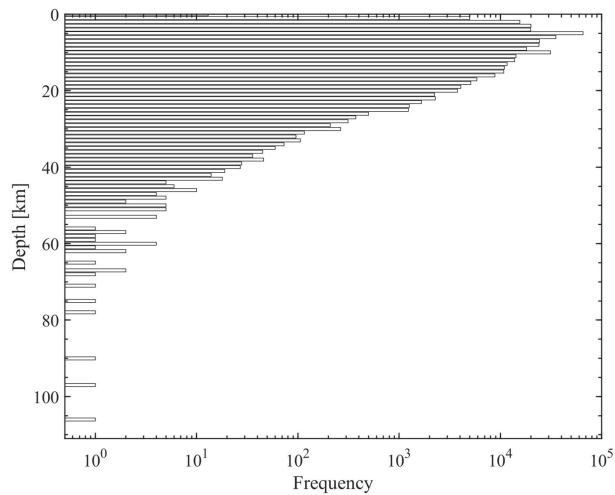


Figure 13. Depth-frequency distribution of earthquakes in CSES.

Acknowledgement

The earthquake catalogs in this report are sourced from the China Earthquake Networks Center (CENC) through the internal link provided by the Earthquake Cataloging System at the China Earthquake Administration, available at <http://10.5.160.18/console/index.action> (last accessed: December 18, 2023), with a Digital Object Identifier (DOI) of 10.11998/SeisDmc/SN. The Bayesian Magnitude Completeness (BMC) method used for calculating the spatial distribution of M_c was provided by Dr. Arnaud Mignan. The Epidemic Type Aftershock Sequence (ETAS) model code for simulating earthquake catalogs and calculating earthquake probabilities was derived from a code shared by Dr. Shyam Nandan which is used in (Nandan et al., 2021a;b; 2022). Fault data were obtained from the GMT Chinese Community at <https://docs.gmt-china.org/latest/dataset-CN/CN-faults/> (last accessed: May 31, 2024). This work is partially supported by the National Natural Science Foundation of China (Grant no. U2039202 and U2039207), the Guangdong Basic and Applied Basic Research Foundation (Grant No. 2024A1515011568), the Special Fund of the China Seismic Experimental Site 2022 (DQJB22Z01-10), the Shenzhen Science and Technology Innovation Commission (Grant no. GJHZ20210705141805017), and the Center for Computational Science and Engineering at Southern University of Science and Technology.

References

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Attachment: About AoyuX

AoyuX is a developing online research platform dedicated to advancing earthquake predictability in China. It is produced by the Institute of Risk Analysis, Prediction, and Management (RisksX) at the Southern University of Science and Technology (SUSTech), in collaboration with multiple research institutions. In Chinese mythology, "Aoyu" is a legendary creature with a dragon's head and a fish's body, believed by ancient Chinese to cause earthquakes. The "X" represents the unknown and infinite in Western culture, symbolizing goals and hopes.

The AoyuX platform aims to use cutting-edge statistical seismology tools for forward earthquake forecasting research. It will also serve as an online research platform for forward earthquake forecasting competitions with

other models. In the future, the AoyuX platform will integrate geophysical observation data, artificial intelligence (AI), and other technologies to conduct various types of earthquake forecasting research, including statistical, numerical, and physical methods. AoyuX is dedicated to help building a modern earthquake forecasting evaluation system in China, encompassing mid-term, short-term and imminent forecasts. It will develop a new visualization platform for earthquake predictability research, facilitating communication between experts and the public, and serving the combined efforts of experts and the public in tracking and assessing the imminent occurrence of major earthquakes, as well as in mid-term and short-term earthquake forecasting.

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